

# BHARATH BARMA

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## Education

### Clemson University

Aug 2023 – May 2025

*Master of Science in Computer Science GPA:3.8*

### Sathyabama Institute of Science and Technology

Aug 2019 – May 2023

*Bachelors of Technology in Computer Science GPA:4.0*

## Skills & Certifications

**Programming Languages:** Python, Java, JavaScript, TypeScript, C++,HTML ,CSS

**Web Development:** Node.js, Express.js, FastAPI, React.js, Next.js,AngularJS, MYSQL(Query language)

**Cloud & DevOps &Database:** AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, PostgreSQL, Azure, MongoDB

**Software Development Tools:** Git, Postman, RESTful APIs, WebSockets, CI/CD Pipelines, OAuth2 ,SAML,OIDC

**System Design:** Microservices Architecture, API Design, SaaS Deployment,Terraform ,DataDog, PagerDuty,Redis

**Certifications:** AWS Certified Solutions Architect – Associate

## Experience

### Research Assistant – AI & Full Stack Development

May 2024 – Apr 2025

*Clemson University*

*Clemson, South Carolina*

- **Conducted** research in **educational data mining**, analyzing **1M+** student behavior logs to identify learning patterns and generate predictive insights with **95%+ accuracy**, enabling personalized learning strategies and boosting student engagement by **20%**.
- **Developed** secure **RESTful APIs** using **Node.js (Express.js)** and **Python (Flask)** to collect and process **500+** student learning session records, ensuring data integrity and supporting large-scale research analysis.
- **Built** responsive web applications with **React.js**, integrating interactive charts and connecting to **Node.js APIs** for real-time data updates, improving research team efficiency by **30%**.
- **Enhanced** accessibility and performance by applying **Tailwind CSS** and modular design principles, cutting page load times by **30%** and ensuring compliance with accessibility standards.
- **Designed** and optimized **PostgreSQL** schema with **20+** indexed tables, improving query performance by **40%** under high-volume ingestion workloads and ensuring scalability for future research.
- **Automated** deployment and monitoring pipelines with **Docker** and **GitHub Actions**, achieving **99.9% uptime** on **AWS EC2** and reducing release errors by **25%**, while enabling continuous delivery of research features.

### Smart Internz

Nov 2022 – July 2023

*Software Development Engineer Intern*

*Chennai, India*

- **Developed** a **MERN stack** faculty feedback system allowing used by 300+ students to submit anonymous feedback,using **React** for form interfaces and **Node.js** with Express for backend routing.
- Integrated **MongoDB** to store and analyze **2000+** feedback records and implemented aggregation pipelines for generating department-level insights.
- **Built** dashboards with **React.js** and **Chart.js**, enabling real-time data visualization of 5+ key metrics for system monitoring.
- **Enabled** email notifications using **NodeMailer**, sending **100+** alerts to faculty on new feedback submissions while maintaining privacy.
- **Deployed** the application using **Docker** and managed **Git workflows**, reducing deployment errors by **40%** and ensuring environment consistency.

## Projects

### ALL AI- Saas Project

- **Built** an **AI SaaS** platform using **React**, **Tailwind CSS**, **Node.js**, **Express.js**, and with **PostgreSQL** backend.
- Integrated **OpenAI**, **Gemini**, **cloudinary** APIs for generating images, blogs ,and removing objects in real-time.
- Used **Clerk** for authentication, **react-hook-form** for validation, and react-toast for server error handling.
- **Deployed** on **Vercel** and **GCP** enabling fast load times, modular design, and secure **API routing**.

### Personal Finance Tracker

- **Built** a personal finance tool using **React.js** and **Flask** for users to track income, expenses, and savings goals.
- **Integrated PostgreSQL** to store and query 10,000+ transaction records efficiently
- **Designed** data visualizations with **Chart.js** to display trends in spending and budgeting.
- **Deployed** the application on **Heroku** with **GitHub Actions CI/CD** for seamless updates.